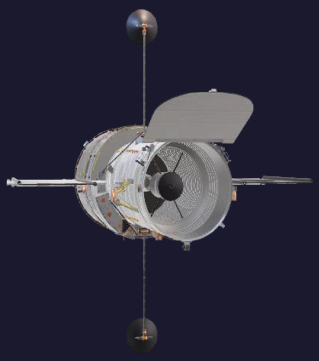
Challenger Space Shuttle Disaster

Team Ethos

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Agenda

- Introduction
- The accident at a glance
- Why it happened?
- Day before the disaster
- Aftermath
- Ethical dilemmas
- Ways this occurrence could have been avoided
- Ethical framework
- Stakeholders
- Decisions and its impacts
- Alternative decisions
- Feynman's Finding
- Lessons learned
- Conclusion

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Introduction





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- The Space Shuttle Challenger disintegrated 73 seconds into its mission on January 28, 1986, killing all seven crew members aboard.
- At 11:39 a.m. EST, the spacecraft destroyed 46,000 feet (14 km) above the Atlantic Ocean, off the coast of Cape Canaveral, Florida (16:39 UTC). It was the first tragic accident involving a flying American spacecraft.
- The mission, designated STS-51-L, was the orbiter's tenth flight and the Space Shuttle fleet's twenty-fifth voyage.
- In addition to bringing schoolteacher Christa McAuliffe into space, the crew planned to install a communications satellite and research Halley's Comet while in orbit.
- The latter produced greater media attention and coverage of the mission than normal.

The Crew

- STS-51-L crew:
- (Backrow) Onizuka, McAuliffe., Jarvis, Resnik;
- (Front row) Smith, Scobee, Ronald Erwin McNair



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Challenger accident at a glance

Launch	• On January 28th 1986, 11:38 am EST
1.0 secs	• Within 1.0sec fumes were observed due to the leaking O-Ring
59 secs	• Visible flame from O-ring
64 secs	• External fuel tank breached
72 secs	• booster begins to rotate and fuel tank collapses due to structural failures
73 secs	• explosion and destruction of STS. Traveling Mach 1.92 at 46,000 ft (14,000 m) altitude at that time
110 secs	• The Range Safety Officer activated the auto-destruct systems on both SRBs 110 s after launch (36 s after explosion).

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Why did the accident happened?

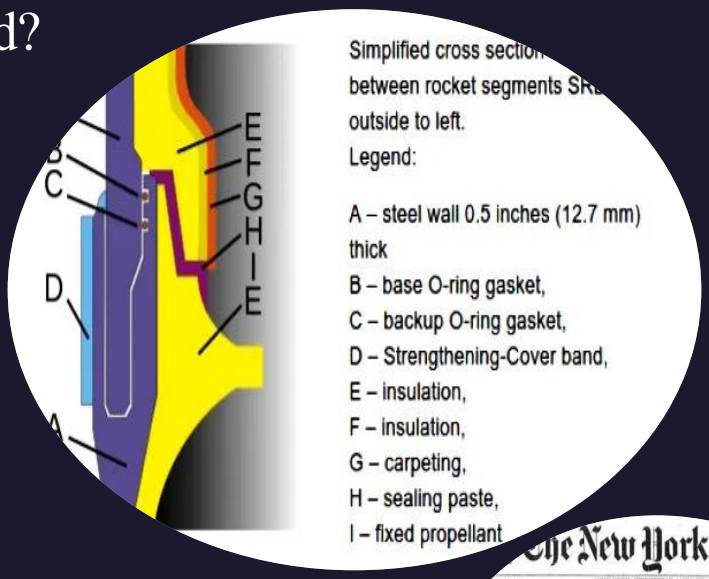
- O-Ring failure :

The major cause of the accident is an O-ring sealing joint failure in the right SRB. Because of the cold temperature, the O-Ring did not expand enough to maintain the seal tight, enabling hot combustion gases to fly through the gap and damage the O-rings.

- Breach in liquid propellant tanks
- No escape systems

It was quite feasible that an abort mechanism might have rescued the astronauts and civilians. This is an example of failing to prioritize safety. Abort systems were installed on every prior and later space shuttle flights.

- Considering this as an acceptable risk





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The Day before the disaster

- Thiokol asked for a teleconference with NASA on January 27 to discuss the issue of O-ring sealing in cold temperature
- The teleconference started at 8:45 p.m. EST and connected NASA Marshal Space Flight Center (MSFC) officials in Alabama and Florida with Morton-Thiokol in Utah.
- Thiokol engineers recommended not to launch unless air temp was at least 53 ˚F , while the predicted temperature at launch was 29 ˚F
- A MSFC project manager responded by saying “When do you want me to launch, next April?”
- Thiokol engineers admitted they lacked statistical analysis to show the correlation of Temperature and O-ring damage .
- Thiokol VP said, “Take off your engineering hat and put on your management hat.”
- The Telecon continued at 11:00 p.m., with Thiokol VP stating that the data was inconclusive and that the corporation advocated launching.
- Al McDonald refused to sign the launch authorization on Thiokol's behalf, therefore a Thiokol official in Utah had to sign off on the launch and fax the authorization to Florida.

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Aftermath

- Morton Thiokol formed a failure investigation team on 31st January 1986 to study the challenger explosion.
- The Rogers Commission Report was authored by a Presidential Commission tasked with examining the Space Shuttle Challenger catastrophe on its tenth mission, STS-51-L. The report, which was released and submitted to President Ronald Reagan on June 9, 1986, determined the cause of the disaster and urged NASA to improve and install new safety features on the shuttles as well as in its organizational handling of future missions.
- In the wake of what happened with Challenger, NASA made technical changes to the shuttle and worked to change the safety and accountability culture of its workforce.
- The shuttle program resumed flights in 1988.

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➢ LEGAL IMPACT

- The engineer, Roger Boisjoly, of Morton Thiokol accused Morton Thiokol of fraud, negligence, manslaughter, racketeering, defamatory statements and untruthful testimony to a Presidential investigating commission and Congress, among other wrongdoings. Plan for $10 Million Claim.
- The Government and Morton Thiokol paid $7,735,000 in cash and annuities as settlement to the families of four of the crew members killed in the Challenger disaster.
- Morton Thiokol, paid $4,641,000 and $3,094,000 was the government's portion of the settlements.

ID: 229035349

➢ ECONOMIC IMPACT

- The challenger shuttle was replaced by endeavour at an expenses of $2 Billion.
- The financial consequences of the event may be divided into two groups. The first, "reconstitution" expenditures, comprise the investigation's fees, any shuttle system adjustments recommended by the study, and the replacement of equipment damaged in the accident. The second category of consequences comprises the expenditures and savings incurred by NASA as a result of the Challenger's demise.
- Reconstitution costs are about $500 million from preliminary NASA estimates.

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➢ SOCIAL IMPACT

- NASA's image suffered a massive blow and the space program was halted for several months after the disaster. It took years to gain the governments and taxpayers trust back in the program.
- More than 60% of the children who watched the explosion of the shuttle on TV feared at least one stimulus related to Challenger within the first 5-7 weeks of the disaster, few suffered from PTSD.
- Although latency-age children's altered attitude about space occupations remained mostly stable, teenagers' reduced expectations for the future in general grew.

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Ethical dilemmas

- Competition between individual space centers: Unforeseen competition from the European Space Agency put NASA in a position where it would have to fly the shuttle dependably on an over-ambitious schedule in order to prove the Space Transportation System's cost-effectiveness and potential for commercialization.
- Pressure to achieve target : NASA managers were anxious to launch the Challenger for several reasons, including economic considerations, political pressures, and scheduling backlogs.
- People fearing opposing the NASA officials.
- Crew being wrongly informed about the risks involved.
- Media pressure.
- Boisjoly taking home the Thiokol documents relating to the disaster.

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Ways it could have been avoided

- Engineers should have gone outside the chain of command to inform higher-level NASA management.
- Proper escape options for the crew
- Proper joint seal design on the SRB



- In the Presidential Commission report: Chairman Rogers raised the question of whether any astronaut office representative was aware [of the O-ring problem]'. Weitz, [an astronaut's representative] answered: "We were not aware of any concern with the O-rings, let alone the effect of weather on the O-rings. "
- A phone call could have prevented the accident!!
- Jesse Moore, NASA's Associate Administrator for Space Flight and Gene Thomas, the Launch Director were kept in dark regarding the meeting at the Thikol factory in Utah, in which the engineers opposed the launch of the space shuttle.

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- Either one of them, Moore or Thomas, could and would have postponed the flight, if they had had that information.
- While the manufacturer did not tell NASA not to launch, it warned that the impact of launchpad ice on the shuttle was an unknown condition, and that risk of ice striking the shuttle was a potential flight safety hazard. Yet NASA proceeded
- Bob Ebeling was one of the engineers from Morton Thiokol who tried to stop the Challenger's launch outright .
- Escape options could have been made for the shuttle crew, the five astronauts and two civilians on board were not informed of the known risk i.e, about the faulty O-rings they were taking.
- Re-designing the joint seals on the SRB could have been done.
- Nevertheless, there was another option for how the case could have progressed. Better decision-making from the management could have prevented the explosion from happening. For example, the management could order further testing and delay the launch until the trial has been finished. Additionally, the managers could have cancelled the launch to fix the faults in the design of the shuttle, thus also avoiding the explosion.

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Ethical framework

- Code of ethics laid by National Society of Professional Engineers applicable in this situation-

•1. Hold paramount the safety, health, and welfare of the public.
•3. Issue public statements only in an objective and truthful manner.
•5. Avoid deceptive acts.
•6. Conduct themselves honorably, responsibly, ethically, and lawfully so as to enhance the honor, reputation, and usefulness of the profession.

- Engineers are expected under the ASME Code of Ethics to prioritise public safety, health, and welfare in carrying out their professional responsibilities. Every major engineering code of ethics stresses the need of engineers putting public safety and well-being first..

- Although company loyalty is crucial, it should not trump the engineer's civic obligation. Marcia Baron states in her great book on loyalty: "Commitment produces obstinacy, which is a terrible truth. Pursuing a goal with zeal may be deeply romantic, even motivating. It is, however, hard to argue for engineers, who have such a large effect on public safety. Irresponsibility, whether driven by avarice or gloriously unselfish zeal, may be deadly."

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Identify stakeholders

- Marshall Space Flight Center in Huntsville, Alabama, and contractor Morton Thiokol, Inc., in Ogden, Utah
- Marshall was responsible for the shuttle boosters, engines, and tank, while Morton Thiokol manufactured the booster motors and assembled them at the Kennedy Space Center at Cape Canaveral, Florida.
- Lockheed Martin had run structural tests for the shuttle
- Taxpayers
- US Government

Decisions and its Impacts

- NASA was under constant pressure from the government to keep its budget low and its operations highly predictable. Delays were frowned upon, and cost overruns were scrutinized more than safety and mission success.
- NASA staff opposed the delay. “My God, Thiokol, When do you want me to launch — next April?” said Lawrence Mulloy, one of the shuttle program manager attending the teleconference.
- With the pressure from NASA, Thiokol management gave their approval to the launch, and Challenger was on its way to disaster. This failure in communication, combined with a management structure that allowed NASA to bypass safety requirements, was the organizational cause of the Challenger disaster.

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Alternative decisions

- After months of inquiry, it became evident that a single phone call might have avoided the catastrophe. It might have gone to either Jesse Moore, NASA's Associate Administrator for Space Flight, or Gene Thomas, the Launch Director, that morning.
- Better managerial decision-making may have prevented the explosion from occurring. For example, management might request further testing and postpone the launch until the trial is completed.
- The company should go through a series of reliability and quality assurance functions in the launch decision process.

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Tuesday, February 2, 20XX

Feynman's Findings

- According to Feynman, the Safety Factor of 3 for the O rings as mentioned by NASA had a huge error in it.
- The management's claim that the risk of catastrophic malfunction on the shuttle was 1 in 10^5 was also a risible claim according to him
- At the time of retirement, the Shuttle suffered two catastrophic failures across 135 flights, for a failure rate of 1 in 67.5
- Based on his experiences with NASA's management and engineers, Feynman concluded that the serious deficiencies in NASA management's scientific understanding, the lack of communication between the two camps, and the gross misrepresentation of the Shuttle's dangers, required that NASA take a hiatus from Shuttle launches until it could resolve its internal inconsistencies and present an honest picture of the shuttle's reliability.

ID: 229018765

Tuesday, February 2, 20XX

Lessons learned

- We cannot afford to get complacent.
- When we observe anything that makes us feel insecure, we cannot remain mute.
- We must enable individuals to express their concerns without fear of repercussions.
- Whistleblowing in an ethical sense can save lives.
- Managers should be listening to the right kind of decisions, calibrating risk trades relevantly and taking into account the right combination of views when taking decisions.



Conclusion

- NASA had in place a decision structure and related review procedure in the Space Shuttle Program that appeared to have all of the required protections in place to prevent the accident that occurred in the instance of Mission 51-L.
- The program's astonishing string of achievements had strengthened the public's idea of a fail-safe system while masking the truth that undesired risks had been taken nearly from the start. These accomplishments apparently contributed to NASA officials' willingness to continue taking risks, as well as the purportedly changing premise that missions had to be proven hazardous rather than safe.
- The study initiated by the Challenger catastrophe indicated that long-standing issues had not been adequately addressed, and that even earlier successful missions had most likely been ill-advised.

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- The goal of this investigation has been to analyze the human elements that appear to have made ineffectual a decision-making process that, in theory, should have resulted in a different course of action.
- Our findings show that, no matter how carefully crafted a decision structure appears in terms of the sequence of analysis and choice to which it commits decision-makers, its effective utilization is still dependent on the social, psychological, and communicative environment in which responsible party's function. In the instance of Challenger, there is significant evidence that effects from these three domains had an important role in leading to the decision that were made.
- Several opportunities to avert the launch offered themselves in the preceding 20 hours, but on each time, one or more of the forces we identified arose and diminished the odds of changing the crash path that NASA had set itself. A simple act of dissent by a probably overconfident Level III official undermined the respect NASA had earned and must now work to restore.

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Thank You



Tuesday, February 2, 20XX